

Myeongjin Shin

@ hanwoolmj@kaist.ac.kr | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#) | [Daejeon, Korea](#)

EDUCATION

KAIST

B.Sc. in School of Computing, Double major in Mathematical Sciences

Daejeon, Korea

Mar 2021 – Aug 2027

Gyeonggi Science High School For The Gifted

Computer Science

Suwon, Korea

Mar 2018 – Feb 2021

RESEARCH & WORK EXPERIENCE

Research Institute of Mathematics, Seoul National University.

Undergraduate Researcher

Seoul, Korea

Feb 2022 – Present

- **Keyword: Quantum entropies, Variational quantum algorithms, Trace of quantum state powers, Quantum singular value transformation, Hamiltonian certification**
- Conducted theoretical and numerical research on estimation of quantum entropies, quantum distance measures using variational quantum algorithms (VQA).
- Conducted theoretical and numerical research on estimating quantum nonlinear properties. Established a resource efficient algorithm for the estimation of trace of quantum state powers.
- Conducted theoretical research on application of quantum singular value transformation (QSVT). Utilized QSVT to establish the near optimal quantum algorithm for estimating Shannon entropy .
- Conducted theoretical research on Hamiltonian learning. Established an optimal bound for certification of constant-local Hamiltonians with real time evolution.

KAIST Physics Quantum Information Theory Group.

Undergraduate Researcher

Daejeon, Korea

Dec 2025 – Present

- **Keyword: Hamiltonian learning, Decoded quantum interferometry**
- Conducting theoretical research on Hamiltonian learning. Establishing a Hamiltonian learning algorithm with large time resolution.
- Conducting theoretical and numerical research on decoded quantum interferometry (DQI).

KRAFTON

Internship

Seoul, Korea

Jul 2023 – Aug 2023

- Researched AI optimization methods and image processing

Young Inch Lab

CEO

Seoul, Korea

Apr 2022 – Jan 2023

- Managed a startup that enables easier access to subtitles

PUBLICATIONS

^($\alpha-\beta$) Alphabetical order (theoretical computer science convention).

Preprints

Junseo Lee, **Myeongjin Shin**^($\alpha-\beta$), “Optimal certification of constant-local Hamiltonians”, arxiv preprint (Dec 2025)

Myeongjin Shin, Kabgyun Jeong, “Near optimal quantum algorithm for estimating Shannon entropy”, arxiv preprint (Sep 2025)

Journal Articles

Donghwa Ji, Junseo Lee, **Myeongjin Shin**, IlKwon Sohn, Kabgyun Jeong, “Bounding quantum uncommon information with quantum neural estimators”, Quantum Science and Technology 11 (1), 015001 (Nov 2025)

Mingyu Lee, **Myeongjin Shin**, Junseo Lee, Kabgyun Jeong, “Mutual information maximizing quantum generative adversarial networks”, Scientific Reports 15 (1), 32835 (Sep 2025)

Myeongjin Shin, Junseo Lee, Seungwoo Lee, Kabgyun Jeong, “Resource-efficient algorithm for estimating the trace of quantum state powers”, Quantum 9, 1832 (Aug 2025)

Myeongjin Shin, Seungwoo Lee, Junseo Lee, Mingyu Lee, Donghwa Ji, Hyeonjun Yeo, Kabgyun Jeong, “Disentangling quantum neural networks for unified estimation of quantum entropies and distance measures”, Physical Review A 110 (6), 062418 (Dec 2024)

Myeongjin Shin, Junseo Lee, Kabgyun Jeong, “Estimating quantum mutual information through a quantum neural network”, Quantum Information Processing 23 (2), 1-16 (Feb 2024)

RESEARCH TALKS

Optimal certification of constant-local Hamiltonians

[Invited] KAIST Physics QIT Lab Seminar (Dec 2025)

Near optimal quantum algorithm for estimating Shannon entropy

[Invited] Team QST Summer Workshop (Aug 2025)

Resource-efficient algorithm for estimating the trace of quantum state powers

[Invited] Quantum Information Science Club Association (QISCA) (Jun 2025)

[Invited] Electronics & Telecommunications Research Institute (ETRI) (Dec 2024)

[Invited] Seoul National University QST Seminar (Dec 2024)

[Contributed] Annual Meeting of Korean Mathematical Society (KMS) (Oct 2024)

[Poster] Annual Conference on Quantum Information Processing (QIP 2025) (Feb 2025)

Disentanglement provide a unified estimation for quantum entropies and distance measures

[Contributed] Korean Physical Society spring meeting (Apr 2024)

[Contributed] Quantum Information Society of Korea (Apr 2024)

[Contributed] Korean Society for Industrial and Applied Mathematics (Nov 2023)

[Poster] Annual Conference on Quantum Information Processing (QIP 2024) (Jan 2024)

Mutual information maximizing quantum generative adversarial networks

[Invited] North Carolina State University Triangle Quantum Computing Seminar (Nov 2023)

[Contributed] Korean Society for Industrial and Applied Mathematics (Nov 2023)

[Poster] Annual Conference on Quantum Information Processing (QIP 2024) (Jan 2024)

Estimating Quantum Mutual Information Through A Quantum Neural Network

[Invited] National Institute of Science Education and Research Bhubaneswar, India (Aug 2023)

[Contributed] 1rd CQNC Winter Kick-off Workshop (Nov 2022)

[Poster] Asian Quantum Information Science Conference (AQIS 2023) (Aug 2023)

[Poster] Conference on the Theory of Quantum Computation (TQC 2023) (Jul 2023)

PROFESSIONAL ACTIVITIES

Journal Reviewer

Quantum Information Processing Journal (QINP)

TEACHING EXPERIENCE

Quantum Algorithms	Winter 2025
Lectured about quantum decoded interferometry (DQI)	
Quantum Sensing and Hamiltonian Learning Study Group	Fall 2025
Organized a study about quantum computational sensing and Hamiltonian learning with real time evolution	
Quantum Learning and Complexity Theory	Summer 2025
Lectured about random unitaries, classical shadows, quantum property testing, QSVT, QAE, etc	

HONORS

KRAFTON AI Fellowship	Jul 2023 – Aug 2023
KAIST Presidential Fellowship(KPF)	Mar 2021 –

AWARDS & ACHIEVEMENTS

2024 Quantum Hackathon Korea	Jun 2024
1st place. Ministry of Science and ICT prize.	
WorldQuant Brain research consultant	Sep 2023
2023 Quantum Hackathon Korea	Jun 2023
3st place. IonQ special prize.	
KRAFTON AI Fellowship	Jul 2023
10,000k ₩ prize money	
QHack certificate of achievement (advanced)	Feb 2023
Xanadu QHack Coding Challenges	Feb 2023
Power-Up Awards, IBM Quantum/PINQ2 and NVIDIA/Cytxtera/Run :AI	
KAIST Presidential Fellowship(KPF)	Mar 2021
Best Paper Award from Gyeonggi Science High School	Feb 2021
1st place. “Fast elliptic curve operation using dynamic programming on jacobian coordinates(2020)”	

SKILLS AND QUALIFICATIONS

Programming languages & Tools
Advanced skills: C/C++, Python
Basic skills: Qiskit, Pennylane, Pytorch, Tensorflow, JAVA, Javascript, Rust, React, Processing
Algorithms and problem solving skills
Codeforces rating 1894 (Expert), max rating 1962 (Candidate master), godthinkun
Languages
Native Korean
Fluent English (TOEFL iBT 108)
Basic German, Spanish

INTERESTS

- **Quantum Learning Theory:** Hamiltonian learning, testing and certification, Gibbs state learning, Agnostic tomography, Quantum memory-sample complexity tradeoff.
- **Quantum Algorithms:** Applications of quantum singular value transformation (QSVT), Decoded quantum interferometry (DQI).
- **Quantum Property Testing:** (Optimal) estimation of quantum entropies and distance measures, Estimation of nonlinear properties of the quantum state (such as trace of quantum state powers).
- Quantum Error Correction
- Machine Learning and it’s Applications.

Last updated: Dec, 2025.